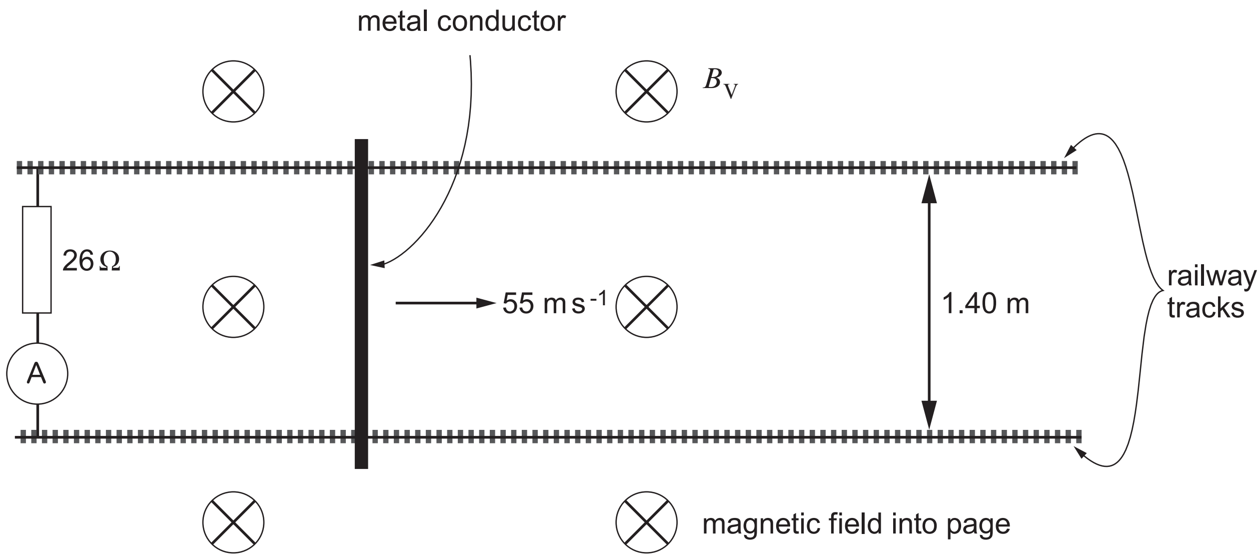
Unit 4 2017.pdf





**5.** A metal conductor is placed across horizontal railway tracks and moved quickly in the direction shown. This set up is designed to measure the vertical component ($B_V$) of the Earth's magnetic field.




*(a)* (i) Explain why a current is detected by the ammeter. [2]

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(ii) State why the current is independent of the horizontal component of the Earth's magnetic field. [1]

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*(b)* Calculate a value for the vertical component of the Earth's magnetic field given that the reading on the ammeter is 184 μA. [3]

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Examiner only

(c) State the direction of the current in the resistor and how you obtained this direction. [2]

(d) A student suggests that the opposing force due to the magnetic field on the moving conductor is negligible compared with other resistive forces. Is the student correct? Justify your answer with a calculation. [4]

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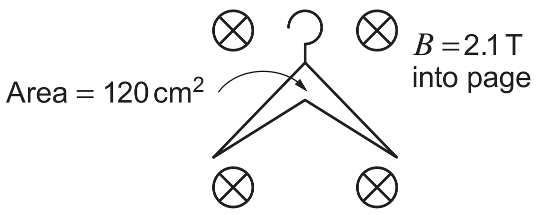
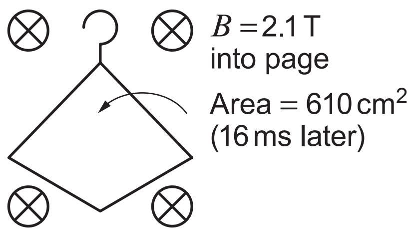
Examiner only

**5.** An experiment is carried out in a very strong uniform magnetic field in order to confirm Faraday's Law under extreme conditions. A coat hanger made of aluminium wire is bent from Shape 1 to Shape 2 in a time of 16 ms.




Shape 1




Shape 2

*(a)* (i) Show that a mean emf of 6.4 V is induced in the coat hanger. [2]

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(ii) Show on the diagram of Shape 2 the direction of the induced current and state very briefly how you determined this direction. [2]

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*(b)* The aluminium wire has a circular cross-section of diameter 3.0 mm and the length of the wire through which the current flows is 91 cm. Show that the mean current in the wire is approximately 1 900 A (resistivity of aluminium = $2.65 \times 10^{-8}\ \Omega$ m). [3]

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Examiner only

(c) Iestyn claims that changing the shape of the coat hanger from Shape 1 to Shape 2 in 16 ms in the magnetic field will increase its temperature by less than 1 °C. Determine, using appropriate calculations, whether or not Iestyn is correct. (Density of aluminium = 2 700 kg m$^{-3}$, specific heat capacity of aluminium = 897 J kg$^{-1}$ K$^{-1}$.) [5]

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(d) For medical research, it is decided to investigate the effect of this strong magnetic field (2.1 T) on patients with metal replacement joints to see if the metal joints become hot or undergo large forces (during MRI scans). Discuss the ethics of such an experiment. [3]

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Unit 4 2019.pdf



Examiner only

**5.** *(a)* State the laws of Faraday and Lenz for electromagnetic induction. [2]

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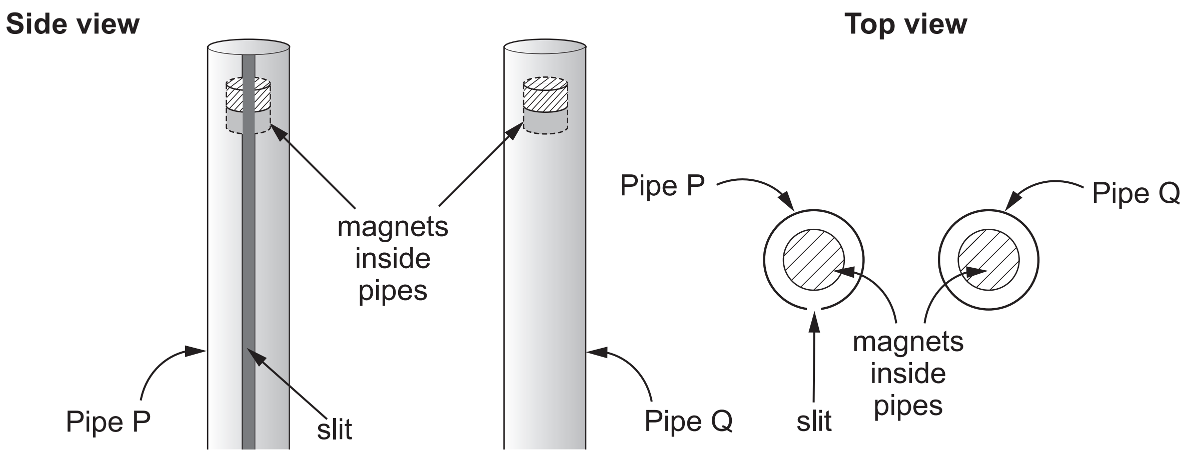
*(b)* Two strong bar magnets are dropped through two copper pipes (P and Q). Pipe P has a slit running along its length but Q is complete. When the magnet is dropped through pipe P it accelerates almost uniformly but the magnet dropped through pipe Q quickly reaches a very low terminal velocity. Explain these observations. [6 QER]

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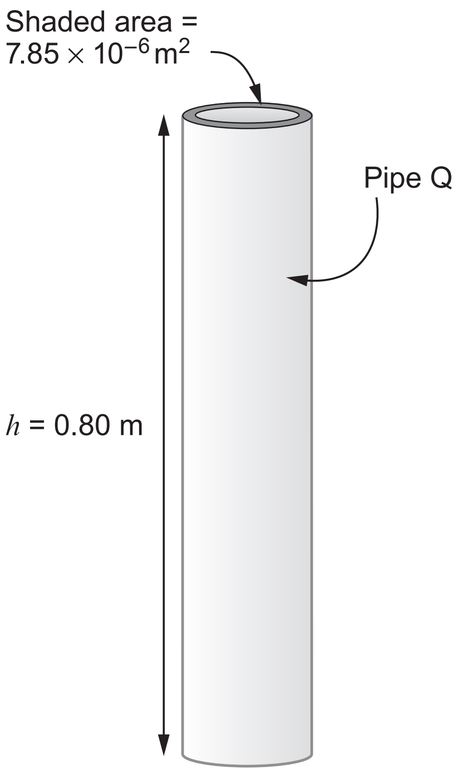
(c) By applying the principle of conservation of energy, calculate the temperature increase of pipe Q after the magnet has fallen at constant speed. [4]

Data:
- mass of magnet = 0.300 kg,
- cross-sectional area of the copper walls of pipe Q = $7.85 \times 10^{-6}\,\text{m}^2$ (see diagram)
- density of copper = $8\,960\,\text{kg}\,\text{m}^{-3}$
- specific heat capacity of copper = $385\,\text{J}\,\text{K}^{-1}\,\text{kg}^{-1}$.

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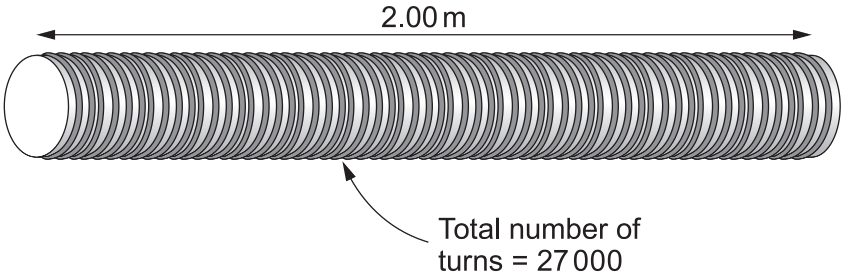
Answer **all** questions.

**1.** (a) (i) Lindsey calculates that a current of 3.57 A will produce a magnetic flux density of 0.121 T inside the long solenoid shown. Determine whether or not she is correct. [3]

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(ii) State how the magnetic flux density inside the solenoid can be increased greatly without changing the current or the number of turns per unit length. [1]

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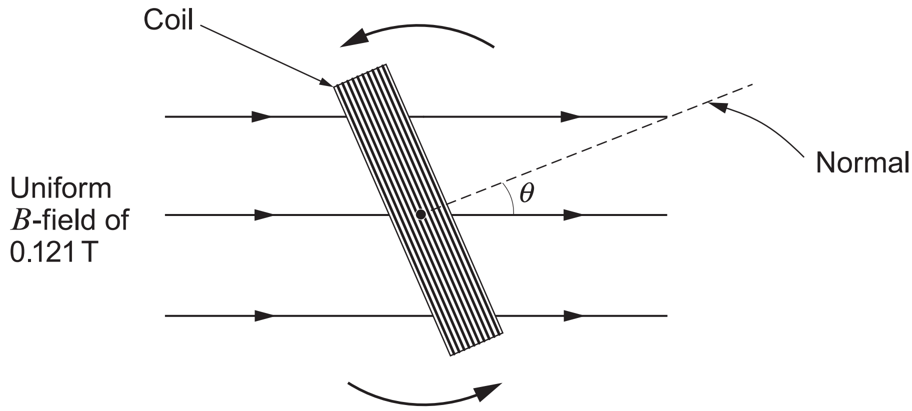
(b) A rectangular coil rotates at a constant angular velocity within a uniform magnetic field of 0.121 T. The coil has 70 turns and cross-sectional area 59 $cm^2$. The diagram below shows the coil, looking along the axis of rotation.




(i) Calculate the flux linkage of the coil when $\theta = 23°$. [2]

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(ii) As the coil rotates, explain what values of $\theta$ provide the maximum **and** minimum values of the induced emf. [2]

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**Turn over.**

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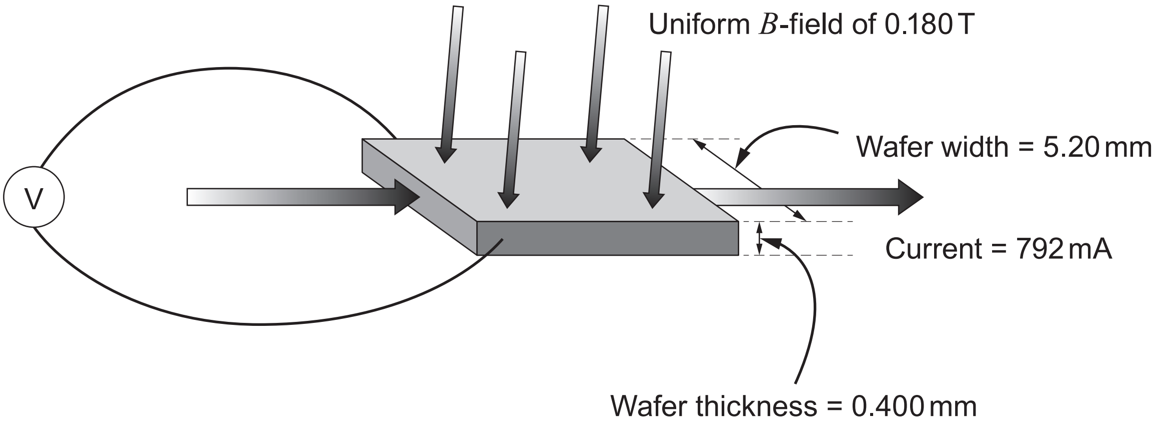
Examiner only

**6.** (a) Catrin carries out a Hall effect experiment to find out the number of free electrons per unit volume in a certain metal. She places a wafer of the metal in a known magnetic field and passes a current through it (see diagram).




(i) By considering the forces acting on free electrons passing through the wafer, explain why a Hall voltage is measured on the voltmeter shown. [4]

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(ii) By equating the magnetic and electrical forces, show that the Hall voltage, $V_H$, is given by:

$$V_H = Bvd$$

where $B$ is the magnetic flux density, $d$ is the width of the wafer and $v$ is the drift velocity. [3]

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(iii) Catrin states that her measured Hall voltage of 68.0 nV is consistent with a drift velocity of approximately 0.07 mm s$^{-1}$. Determine whether, or not, she is correct. [2]

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(iv) Calculate the number of free electrons per unit volume for the metal. [3]

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(b) After repeating the same experiment on a different metal, Catrin obtains a value of $(5.85 \pm 0.19) \times 10^{28}\,\text{m}^{-3}$, for the number of free electrons per unit volume. She is given a table of values in order to determine which metal has been used in the experiment.

| Element | Free electron density / $10^{22}\,\text{cm}^{-3}$ |
|---|---|
| Aluminium | 18.1 |
| Barium | 3.15 |
| Copper | 8.47 |
| Gold | 5.90 |
| Iron | 17.0 |
| Silver | 5.86 |

Explain which metal(s) she should conclude has been used in the experiment. [2]

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**END OF PAPER**

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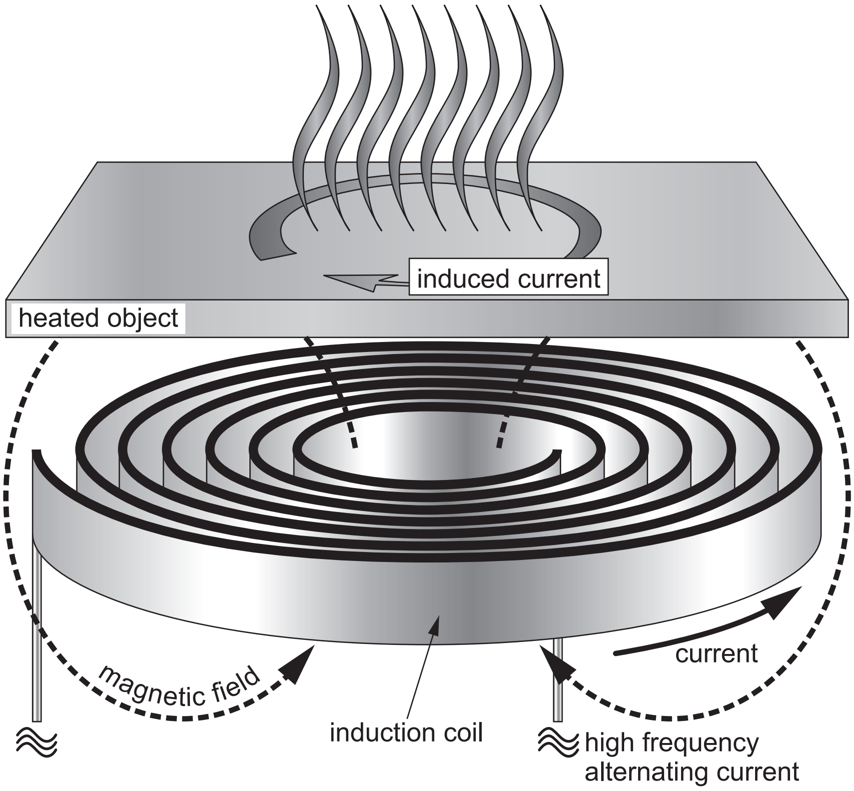
**5.** (a) The diagram shows how conductors can be heated using a varying magnetic field.




Explain why the heated object gets hot **and** why a high frequency alternating current is used. [4]

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(b) Explain how a Hall voltage arises in a Hall probe, how it is measured **and** how this can be used to measure the magnetic flux density. A diagram should be included in your answer. [6 QER]

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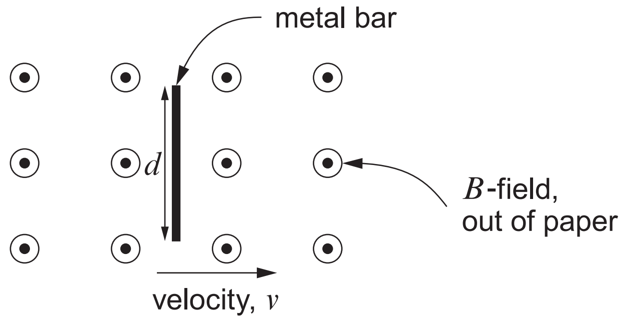
**6.** (a) The diagram shows a metal bar moving with velocity, $v$, through a uniform magnetic field, $B$.




Show that the rate of cutting of flux is $Bvd$. [2]

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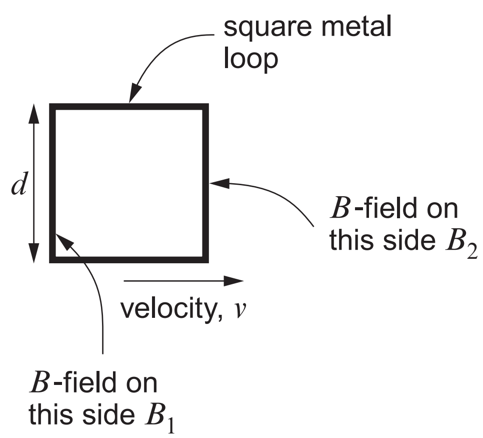
(b) A square metal loop moves through a **non-uniform** $B$-field with the field coming out of the paper.



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(i) Use the expression in part (a) to explain why the induced emf in the square metallic loop is given by: [3]

$$(B_2 - B_1)vd$$

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(ii) The $B$-field, through which the square metal loop moves, increases with distance at a rate of 1.20 T m$^{-1}$ (in the direction of the velocity). The square loop has sides of length 3.8 cm and moves at a speed of 8.8 m s$^{-1}$. Show that the induced emf in the square metal loop is approximately 15 mV. [2]

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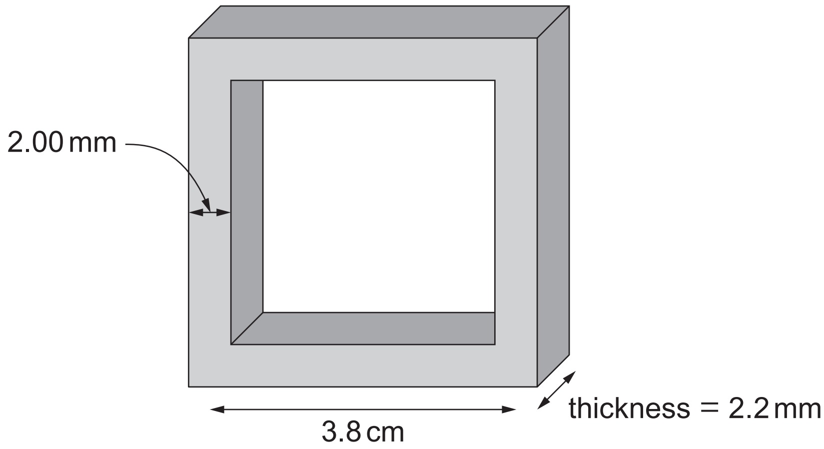
(iii) The square metal loop is made of silver of resistivity $1.59 \times 10^{-8}\,\Omega\,\text{m}$ and its dimensions are shown in the diagram below:




Calculate the current in the square loop. [5]

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